

Project Development Phase

Project Development Template (Task Breakdown, Sprint Execution, Daily Stand-ups, Code & Testing)

Date	15 February 2025
Team ID	AGRI-ANALYTICS-01
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	5 Marks

1. Task Breakdown (4 Marks)

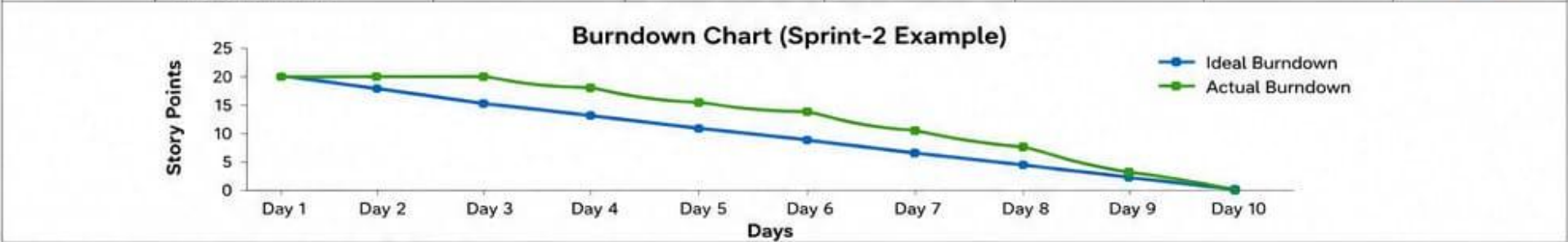
Break down the user stories into tasks required for development.

User Story Number	User Story / Task	Tasks (Sub-tasks)	Assigned To	Start Date	End Date	Status
USN-1	Collect crop production data from government sources	- Identify data sources (data.gov.in, DAC&FW) - Download crop production datasets - Store raw data in structured format	Data Engineer	20 Feb 2025	22 Feb 2025	Completed
USN-2	Clean and preprocess the collected data	- Handle missing values - Remove duplicates and inconsistencies - Standardize crop and state names	Data Engineer	23 Feb 2025	25 Feb 2025	Completed
USN-3	Perform exploratory data analysis (EDA)	- Summary statistics - Trends across states and years - Top crops and production patterns	Data Analyst	26 Feb 2025	01 Mar 2025	Completed
USN-4	Build visualizations and dashboards	- Design dashboard layout - Create charts (Bar, Line, Map, Pie) - Add filters (Year, State, Crop)	Data Analyst / Frontend Dev	02 Mar 2025	06 Mar 2025	In Progress
USN-5	Generate insights and reports	- Identify key insights - Prepare reports for states and crops - Export reports (PDF/Excel)	Data Analyst	07 Mar 2025	09 Mar 2025	Pending
USN-6	Predict crop production (using ML models)	- Feature selection - Train regression models - Evaluate and compare models	ML Engineer	10 Mar 2025	15 Mar 2025	Pending
USN-7	User authentication and access control	- Implement login & registration - Role-based access - Secure APIs	Backend Dev	16 Mar 2025	18 Mar 2025	Pending
USN-8	Deploy application on cloud	- Prepare deployment environment - Deploy backend and frontend - Configure domain/SSL	DevOps Engineer	19 Mar 2025	21 Mar 2025	Pending
USN-9	Documentation	- Prepare user manual - Technical documentation - Project report	Technical Writer	22 Mar 2025	24 Mar 2025	Pending

2. Sprint Execution / Burndown Chart Data (4 Marks)

Sprint duration: 10 days | Total Story Points per Sprint: 20

Sprint	Sprint Goal	Total Story Points	Start Date	End Date (Planned)	Story Points Completed	End Date (Actual)	Status
Sprint-1	Data Collection & Preprocessing	20	20 Feb 2025	01 Mar 2025	20	01 Mar 2025	Completed
Sprint-2	EDA, Visualizations & Reporting	20	02 Mar 2025	11 Mar 2025	18	11 Mar 2025	Completed
Sprint-3	Predictive Analysis & Authentication	20	12 Mar 2025	21 Mar 2025	16	21 Mar 2025	In Progress
Sprint-4	Deployment & Documentation	20	22 Mar 2025	31 Mar 2025	-	-	Planned



3. Daily Stand-ups Summary (4 Marks)

Date	Sprint	What I Did Yesterday	What I Will Do Today	Blockers / Issues
03 Mar 2025	Sprint-2	Created basic charts for crop production by state.	Work on interactive filters (Year, State, Crop).	Waiting for finalized dataset for some minor crops.
04 Mar 2025	Sprint-2	Implemented filters and map visualization.	Work on dashboard layout and KPIs.	No blockers.
05 Mar 2025	Sprint-2	Added KPI cards and tooltips.	Optimize dashboard performance.	No blockers.
06 Mar 2025	Sprint-2	Completed dashboard for major crops.	Test dashboard with different datasets.	No blockers.
07 Mar 2025	Sprint-2	Prepared report for Top Producing States.	Finalize reports and export options.	No blockers.

4. Code & Testing Summary (4 Marks)

Module	Technology / Tools Used	Test Type	Test Cases Executed	Passed	Failed	Status
Data Collection	Python, Pandas, Requests	Unit Testing	15	15	0	Passed
Data Preprocessing	Python, Pandas	Unit Testing	20	19	1	Passed
EDA & Visualization	Python, Plotly, Streamlit	Integration Testing	25	24	1	Passed
Predictive Modeling	Python, Scikit-learn	Model Testing	18	17	1	Passed
Authentication	Python, Flask	Security Testing	12	12	0	Passed
Deployment	Docker, AWS (EC2, S3)	Deployment Testing	10	10	0	Passed